

Math 403 Midterm #1, Spring 2025

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I verify that I have consulted no electronic or human sources except those allowed and that I have in all other ways complied with Duke's code of academic honesty.

[1pt] Signature:- I agree

Due Date: 11:59pm Saturday 15 February 2025

READING ASSIGNMENTS

/100

11. for Thu. 13 February

- [Treil, §6.1–§6.2] Spectral theorem, normal operators

12. for Tue. 18 February

- [Treil, §6.3.1–6.3.2] singular values, positive (semi)definite operators
- [Lax, Appendix 10, Theorem 1, p.334–337] Schur factorization
- [Lax, Chap.8: §Spectral Resolution, §Normal Maps]
- [Lax, Chap.9: §Polar Decomposition–§Singular Value Decomposition, p.169–170]

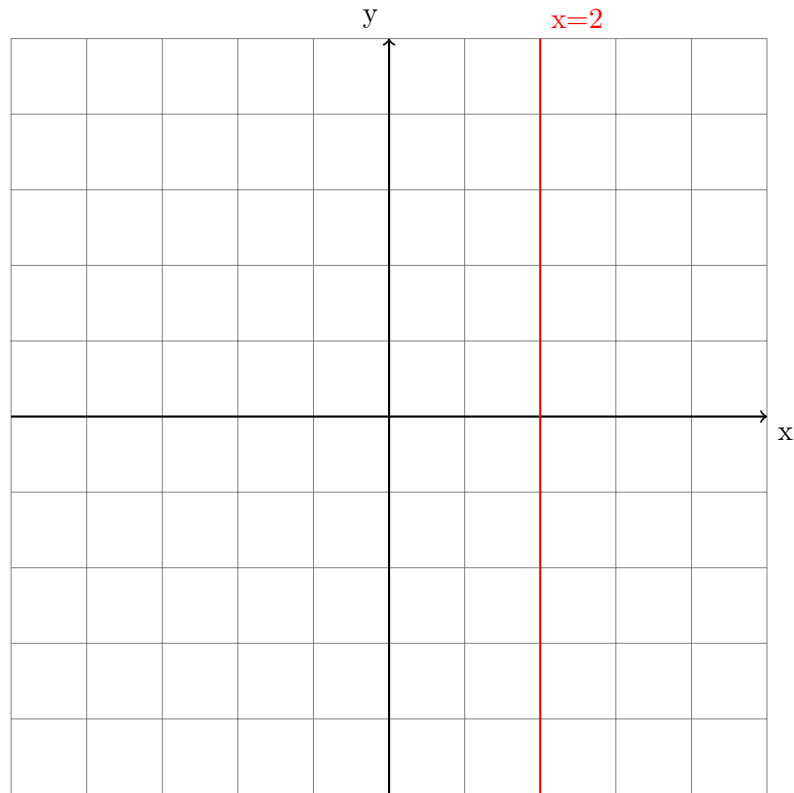
PROBLEMS

1. Let C be a convex cone (Homework #2.11). Prove that 0 is an extreme point of C if [/14](#) the cone C doesn't contain a straight line. Construct a convex cone that contains an entire straight line but has 0 as an extreme point.

Solution

Let's prove the contrapositive, assume that 0 is an extreme point, therefore it is in the middle of the a line segment, between x and y , but since it is a cone, it must be closed under non-negative scaling, therefore you can extend the line segment to infinity on both sides to get a line, which means there is a line if 0 is an extreme point, proving the statement.

The below construction is a counter example to the second part of the question.



All the points on the line $x = 2$ are in the cone and all the non-negative scaling, but 0 isn't an extreme point, because all the points are to the right of the y-axis and therefore the only way to have a line through, 0 is to have a line that is vertical, which is impossible in this case.

2. Fix a finite field $\mathbb{F}_q$ with q elements. Show that q is a power of a prime integer p . Hint: [/14](#)
 $\mathbb{F}_q$ is a vector space over $\mathbb{F}_p$ for some prime p ; why is that true, and how is it useful?

Solution

For notational ease let's denote 2 as $1+1$, 3 as $1+1+1$, and so on. Where 1 is the multiplicative identity in the field. We know that for finite fields the characteristic is a prime number (from homework 1). Another thing worth of note is that the inverses of all the elements listed are also inside this set.

Proof: From number theory we can take Bezout's identity to show that for coprime integers m, n there exists integers a, b such that $am + bn = 1$. In our situation since the field operations can be constructed to resemble integer multiplication and addition under modulo we can see that $\exists n \in \{1, 2, 3, \dots, q-1\}, \exists m \in \mathbb{Z}, \forall q \in \{1, 2, 3, \dots, p-1\}$ such that

$$nq + mp = 1 \Rightarrow nq \equiv 1 \pmod{p} \Rightarrow n^{-1} = q \wedge q \in \{1, 2, 3, \dots, p-1\}$$

Or simply stated each number has modulo inverse for a prime number p . So this set is closed both under addition and multiplication, and therefore constitutes a subfield.

If all the elements in the broader field can be described as a linear combination of 1 we are done since $q = p$, else $\exists g$ such that g is not a linear combination of 1.

We also can't describe $g + 1$ as a linear combination of 1, since that would imply g is a linear combination of 1 because $1, 2, \dots, q-1$ are closed under addition and if $g + 1$ were in this set then $g + 1 + q - 1 = g + 0 = g$ would also be in this set, which is a contradiction. Continuing this argument we can see that $g + 1, g + 2, \dots, g + q - 1$, are all distinct elements of the field. Along the same line of reasoning we can see that $g * 1, g * 2, \dots, g * (q - 1)$ are also not in any of the sets we have seen before because they are all closed under multiplication, therefore each multiple of g can be extended to make p elements and therefore there are in total p^2 elements (corresponding to all the linear combinations of 1 and g) that we have seen so far. We are also left with describing g^2 , but there are really only few cases to consider, one is when g^2 is 1, which is the case in which g is its own inverse, if so then the set we have described so far is closed under multiplication and the hypothesis is correct.

Second is the case when it is not its own inverse, this case is broken up into two cases, one where g^2 can be described as a linear combination of g and 1, in which case the set is still closed, under multiplication, otherwise g^2 is a completely different element than the ones we have seen before.

If g^2 is completely different than the elements we have seen before, then by extension all linear combination of $1, g, g^2$ are also different elements, pushing our size to p^3 , we can progressively increase the number of elements through this procedure until we reach a point when $g^y = 1$. It is inevitable that in a finite field this must happen, because the set of elements is finite and we can't keep generating new elements forever, therefore we must reach a point when $g^y = 1$.

At this point we have either exhausted all the elements in the field, or there is another element x that can't be described as a linear combination of all the elements we have seen so far, but this is analogous to the argument above with the new element being x , and we can keep repeating this argument, each time increasing the number of elements by a power of p , until we reach $q = p^n$.

Note: The main idea of my proof is that each time I encounter a new element that can't be described as a linear combination of the elements I have seen so far, then it must increase the number of elements I have so far, by p . I tried to prove that the set $g, g + 1, g + 2, \dots, g + p - 1$ is closed under multiplication, but I couldn't prove it

unless $g^{-1} = g$, I don't know if this claim is true or not, but I think it is true because when you have a chain of $g * g * g * \dots * g = 1$, you can brake it up in the middle to get $g * g * \dots * g = (g * g * \dots * g)^{-1}$, but this proof would rely on the chain of multiplication having even length, and I don't know if that is true or not for every finite field.

3. Fix a 5×5 matrix A whose sole eigenvalue is λ . Either find the Jordan form of A /14 assuming that $A - \lambda I$, $(A - \lambda I)^2$, and $(A - \lambda I)^3$ have kernels of dimension 2, 3, and 5, respectively, or else prove that no such matrix A exists. Justify your response either way.

Solution

Theorem used:- $\ker(A - \lambda I)^k - \ker(A - \lambda I)^{k-1}$ is the number of jordan blocks of at least size k .

Pf: Prof. Miller said it in class, but I will just prove it here for myself and I can get free feedback on my proof. The dimension of the kernel increases because generalized eigen vectors in each jordan blocks are annihilated in each iteration, and in the each iteration the number of new generalized eigen vectors that get annihilated corresponds to the number of jordan blocks that haven't annihilated all their generalized eigen vectors yet, so for an eigen vector to be annihilated in the k^{th} iteration, it must be in the kernel of $(A - \lambda I)^k$ but not in the kernel of $(A - \lambda I)^{k-1}$. Therefore the size of the jordan block it corresponds to, must be atleast be size k .

Since the kernel of $A - \lambda I$ is 2, then there are 2 jordan blocks (All jordan blocks have size atleast 1). Since the kernel of $(A - \lambda I)^2$ is 3, then only 1 jordan block has size > 1 because if both had size > 1 , then the dimension kernel would increase by 2. So now we have that there are 2 jordan blocks, one of size 1 and one of size at least 2. Since the sum the sizes of all jordan blocks must sum upto 5, then the size of the second jordan block must be 4.

We hit the contradiction because the kernel of $(A - \lambda I)^3$ is 5. The increase in the dimension of the kernel is 2 but we already know that there is only one jordan block that is contributing additional dimensions to the kernel, and it can't contribute more than 1 dimension to the kernel, therefore no such matrix exists (A matrix can't have 2 jordan blocks of size atleast 3 and 1 of size atleast 2).

4. Suppose $\varphi : V \rightarrow W$ is an invertible linear transformation of Hermitian inner product spaces. Prove or construct a counterexample: if $\mathbf{e}_1, \dots, \mathbf{e}_n$ is an orthogonal basis of V and $\|\varphi(\mathbf{e}_i)\| = \|\mathbf{e}_i\|$ for all i , then φ is an isometry: $\|\varphi(\mathbf{v}) - \varphi(\mathbf{v}')\| = \|\mathbf{v} - \mathbf{v}'\|$. [/14](#)

Solution

Counter Example

The matrix $\begin{bmatrix} 1/2 & \sqrt{3}/2 \\ \sqrt{3}/2 & 1/2 \end{bmatrix}$ satisfies the condition above because the image of the basis vectors has unit length but the norm isn't conserved $\sqrt{(\sqrt{3}/2 - 1/2)^2 + (\sqrt{3}/2 - 1/2)^2} \neq 1$.

5. Fix vector spaces $W \subseteq V$ over a field F . Let $T : V \rightarrow V$ be a linear transformation such that $T(W) \subseteq W$. Using the universal property of quotients, prove that T induces a linear transformation $\bar{T} : V/W \rightarrow V/W$ given by $\bar{T}(v + W) = T(v) + W$. Show further that $\bar{T}$ is an isomorphism if T is an isomorphism and V has finite dimension. Must the same hold if V does not have finite dimension? Prove or give a counterexample. /14

Solution

All elements in V can be expressed as $w + v$ where $w \in W$ and $v \in V/W$. Therefore $\bar{T} : V \rightarrow V/W$ defined by $\bar{T}(W + v) = T(v) + W$ is well defined, and its kernel is W , therefore by the universal property of quotients there exists a linear transformation $\bar{T} : V/W \rightarrow V/W$ such that $\bar{T}(v + W) = T(v) + W$.

If T is an isomorphism, then $\bar{T}(v_1 + W) = \bar{T}(v_2 + W) \rightarrow T(v_1) + W = T(v_2) + W \rightarrow T(v_1) = T(v_2)$, the value of w doesn't matter, therefore the two elements in the codimension(V/W) are the same, therefore $\bar{T}$ is injective and since the dimension of both domain and output spaces are the same, it is also surjective, therefore it is an isomorphism.

This proof doesn't work for Infinite dimensional case, because you need not have $T(W) = W$ (The assumption that w doesn't matter isn't valid in the infinite case, because it is not guaranteed that the preimage of w is in W). For example assume you have a the set of basis $e_i \mid i \geq 0, i \in \mathbb{Z}$. Define W as the subspace spanned by the basis vector $e_i, i \equiv 0 \pmod{2}$. Then V/W is spanned by the odd standard vectors.

Define $T(v)$ that maps e_i to e_{i+2} if i is even; e_1 to e_0 ; and everything else to e_{i-2} . T satisfies all the properties ($T(W) \subseteq W$) and is an isomorphism but $\bar{T}$ is not an isomorphism, because e_1 is mapped to 0, therefore $\bar{T}$ has a non-empty kernel, therefore the statement doesn't hold in an infinite dimensional case.

6. Fix integers $0 < d < e < n$. Express $\{\text{chains } V_d \subseteq V_e\}$ of subspaces of F^n of dimensions d and e as a quotient of $\text{GL}_n(F)$ modulo a subgroup. How many such chains are there if $F = \mathbb{F}_q$ is a field with q elements? /14

Solution

The set of chains can be expressed as the quotient of $\text{GL}_n(F)$ modulo the subgroup that fix V_d and V_e . This subgroup is R_d^e . Where R_d^e is the set of matrices where, there is an invertible $d \times d$ matrix in the top left corner of the $n \times n$ matrix, an invertible $(e - d) \times (e - d)$ matrix in the middle ("starting from the $d+1$ row and column and ending at the e row and column"), followed by an $(n - e) \times (n - e)$ invertible matrix in the bottom right corner, and everything else is 0.

$$R_d^e = \begin{bmatrix} A & 0 & 0 \\ 0 & B & 0 \\ 0 & 0 & C \end{bmatrix} \quad \text{Where } A \text{ is } d \times d, B \text{ is } (e - d) \times (e - d), C \text{ is } (n - e) \times (n - e)$$

$$\text{GL}_n(F)/R_d^e = \{\text{Chains } V_d \subseteq V_e\}$$

Why is this true? Because multiplication by R_d^e from the right maintains the subspace of the left d columns and the left e columns, as well, therefore V_d and V_e stay constant, where V_d is the span of the first d columns and V_e is the span of the first e columns. So you can make a bijection between the set of chains and the set of cosets of R_d^e in $\text{GL}_n(F)$, because each coset only describes one chain and each chain can be described by a coset.

From the last homework we know that the number of invertible matrices of size n in a finite field of size q are $\prod_{i=0}^{n-1} (q^n - q^i)$. So we get $\frac{\prod_{i=0}^{n-1} (q^n - q^i)}{\prod_{i=0}^{d-1} (q^d - q^i) \cdot \prod_{i=0}^{e-d-1} (q^{e-d} - q^i) \cdot \prod_{i=0}^{n-e-1} (q^{n-e} - q^i)}$.

7. Is the ∞ -norm $\|\cdot\|_\infty$ on $\mathbb{R}^n$ induced by an inner product? Justify your response. /14

Solution

No, there is no such inner product that induces the infinity norm.

Proof: Assume that there is some norm v that induces the infinity norm on $n = 2$.

$$v(x, x) = \max(|x_1|, |x_2|)$$

Let's find a contradiction with the norm of $(3, 2)$.

$$\begin{aligned} v((3, 2), (3, 2)) &= \max(|3|, |2|) = 3 \\ v((3, 2), (3, 2)) &= v((2, 0), (3, 2)) + v((1, 2), (3, 2)) \quad \text{By Linearity} \\ &= 2 * v((2, 0), (2, 0)) + 2 * v((1, 2), (1, 2)) = 4 + 4 \neq 3 \end{aligned}$$

Since we didn't assume anything about v then the contradiction exists for any function that is an inner product, therefore there is no inner product that induces the infinity norm.